

**Sri Lanka Institute of Information
Technology
Faculty of Computing**



EduFrame

Web-Based Video Browsing System

Web-Based Platform for Managing and Delivering Educational Video Content

Group – Y2-S1-MLB-B4G1-06

Year 2, Semester 1 – 2026

Design Document

IT Number	Name
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IT25103702	Bandara W. G. M. D.
IT25101767	Withana T. T.
IT25103673	Wickramathunga P. P.
IT25100859	Adikari A. M. T. D.

1. Agile Sprint Summaries

Our team has split the development lifecycle of the **Web Based Video Browsing System** into **four distinct Sprints** spanning from Week 5 to Week 12.

1.1 Completed Sprints

Sprint 01 Summary (Weeks 5 – 6)

Focus: Core Infrastructure, Secure Authentication, and Basic Media Discovery [6].

Key Deliverables & Completed Activities:

- **User Provisioning:** Designed responsive user interface forms, built backend API registration endpoints with secure password hashing logic, and established the core database schemas for handling user profiles and access role.
- **Access Control:** Created a secure login portal backed by JSON Web Token (JWT) session generation and routed authentications seamlessly via role-based dashboard redirection.
- **Video Ingestion Engine:** Created a streaming-optimized video file upload interface equipped with automatic size, extension (.mp4), and duration validation rules.
- **Content Discovery:** Developed database endpoints to search and filter published content tags, categories, or titles from a centralized search bar interface.

Sprint 02 Summary (Weeks 7 – 8)

Focus: Academic Assessment Engine and Platform Content Moderation.

Key Deliverables & Completed Activities:

- **Examination Subsystem:** Implemented a full CRUD interface for question banking and custom test timers, coupled with an interactive student evaluation panel featuring active count-down clocks.
- **Automated Evaluation:** Configured a grading calculation module running alongside a PDF certificate generation engine to reward students instantly upon passing tests.
- **Community Watch tools:** Embedded content flagging dialogue boxes onto streaming pages to pipe community reports directly into a private operational quarantine vault.
- **Moderation Dashboard:** Formatted administrative workspaces allowing personnel to approve, reject, or quarantine media, powered by automated string-matching filters to cross-reference banned keywords.

Sprint 03 Summary (Weeks 9 – 10)

Focus: Communication Management, Scheduling Logistics, and Operational Ticketing.

Key Deliverables & Completed Activities:

- **Communication Hub:** Deployed user interfaces and backend REST APIs for creating time-bound announcements and scheduling academic events.
- **Automated Maintenance:** Programmed backend automated cleanup hooks to handle the scheduled removal of expired event entries seamlessly.
- **Alert Dissemination:** Built a dynamic notification broadcast engine to broadcast alerts directly onto student dashboards and calendar widgets.
- **Support Pipeline:** Formulated technical ticket submission paths allowing students and teachers to log platform issues and view real-time status updates.
- **Queue Operations:** Built an admin routing queue featuring status filtering controls, backend resolution endpoints, and status change system alerts.

1.2 Upcoming Sprints

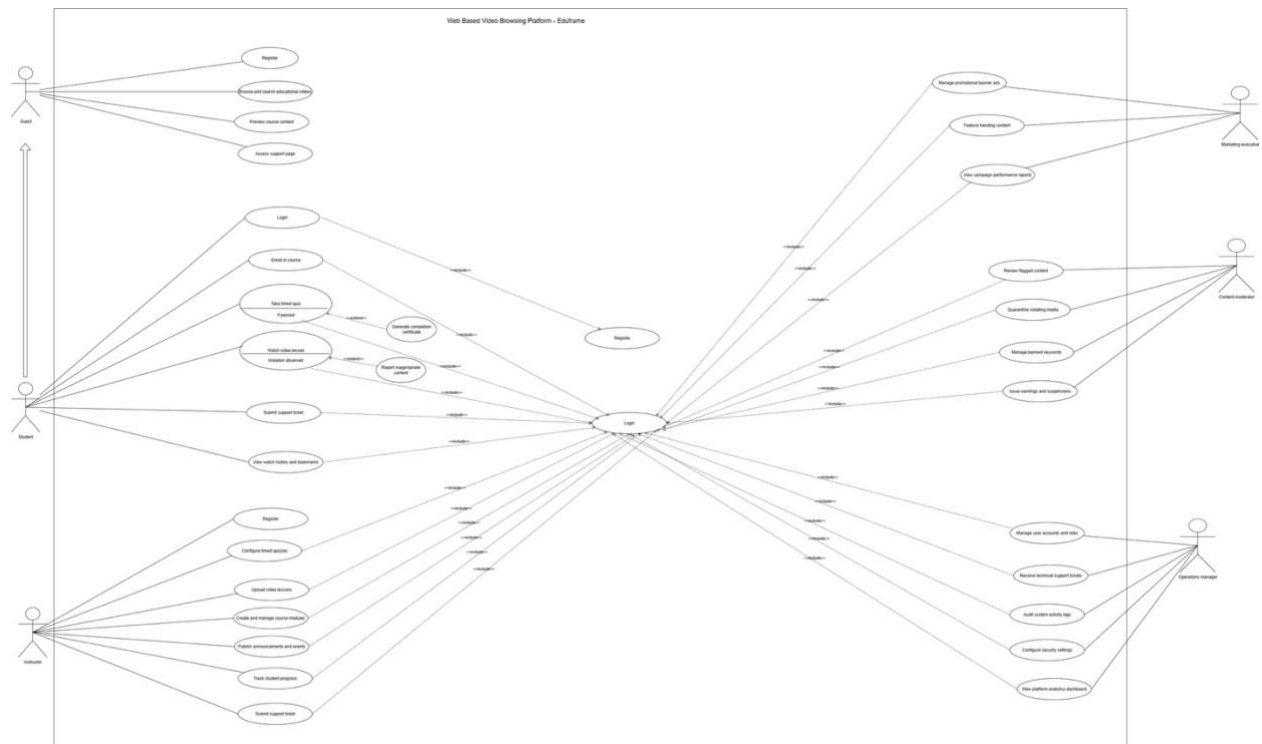
Sprint 04 Summary (Weeks 11 – 12)

Focus: Promotional Infrastructure, Aggregation Dashboards, and System Integration.

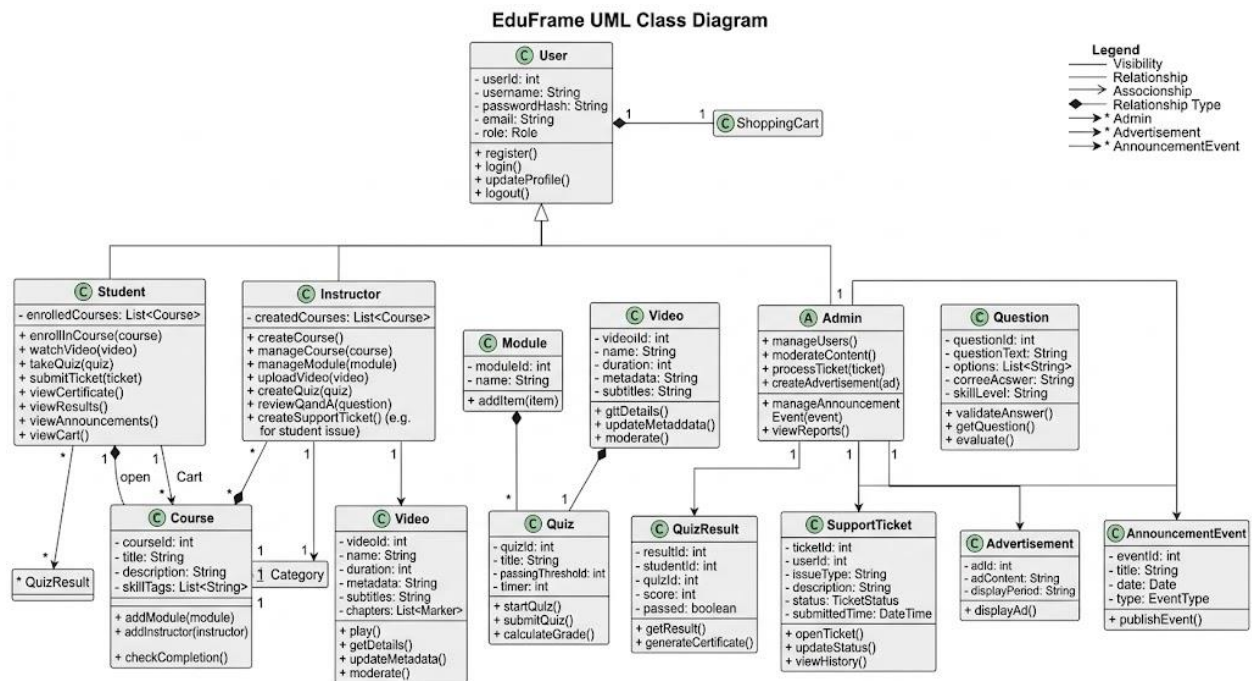
Target Key Deliverables & Expected Activities:

- **Ad Campaign Architecture:** Set up a promotion portal allowing marketing staff to build dynamic homepage banner sliders with time-bound visibility triggers.
- **Analytics Aggregation:** Build high-performance database grouping queries to track view trends, file ingest metrics, and platform performance variables on an admin analytics dashboard.
- **Identity Governance:** Create centralized account tables that let operators toggle access privileges or suspend violating profiles.
- **Full System Verification:** Conduct cross-module end-to-end user path testing and code document cleanups to eliminate technical debt ahead of the Week 13 live demonstration.

2. Use Case Diagram

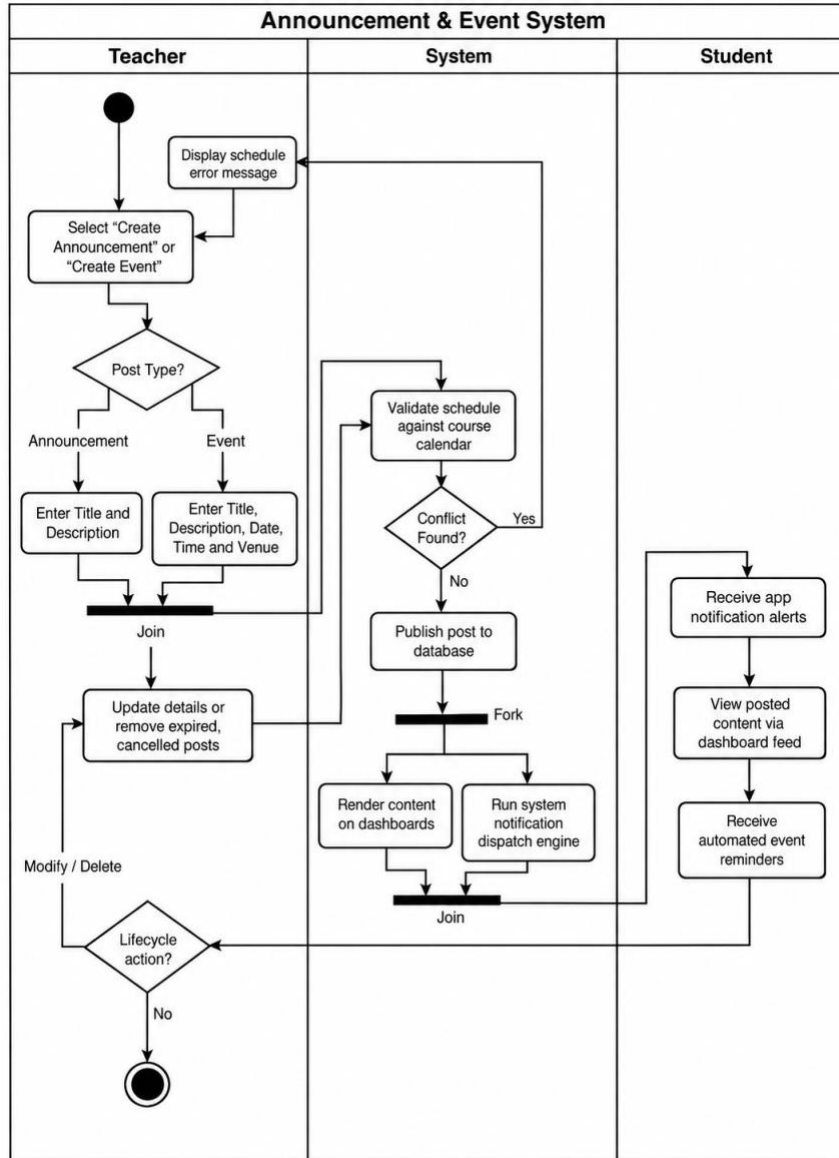


3. Class Diagram

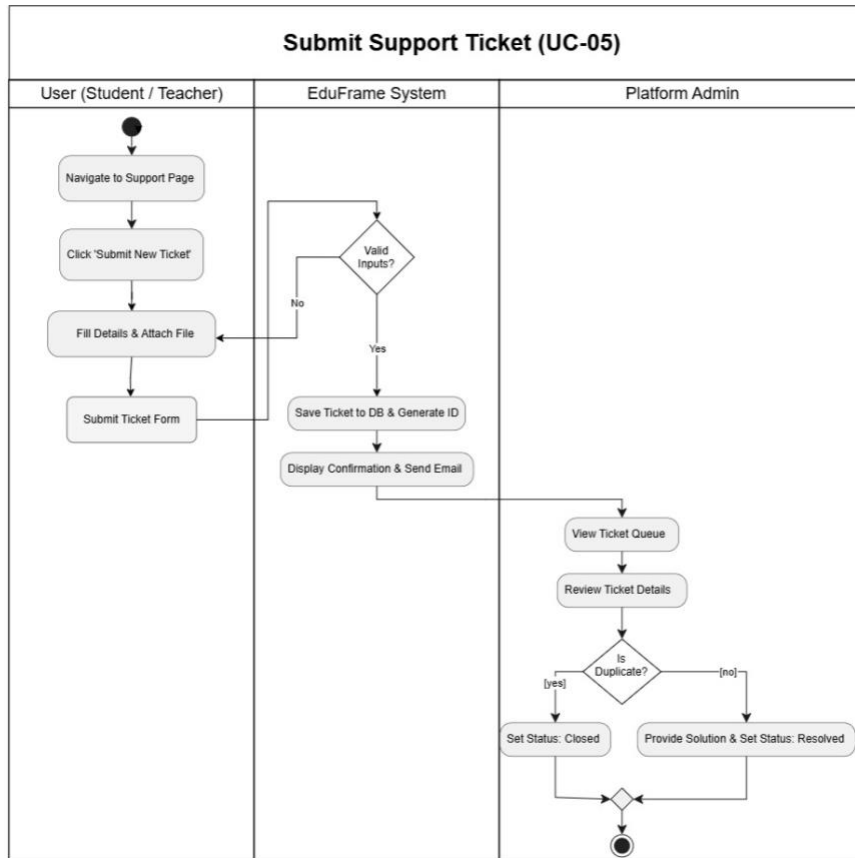


4. Activity Diagrams

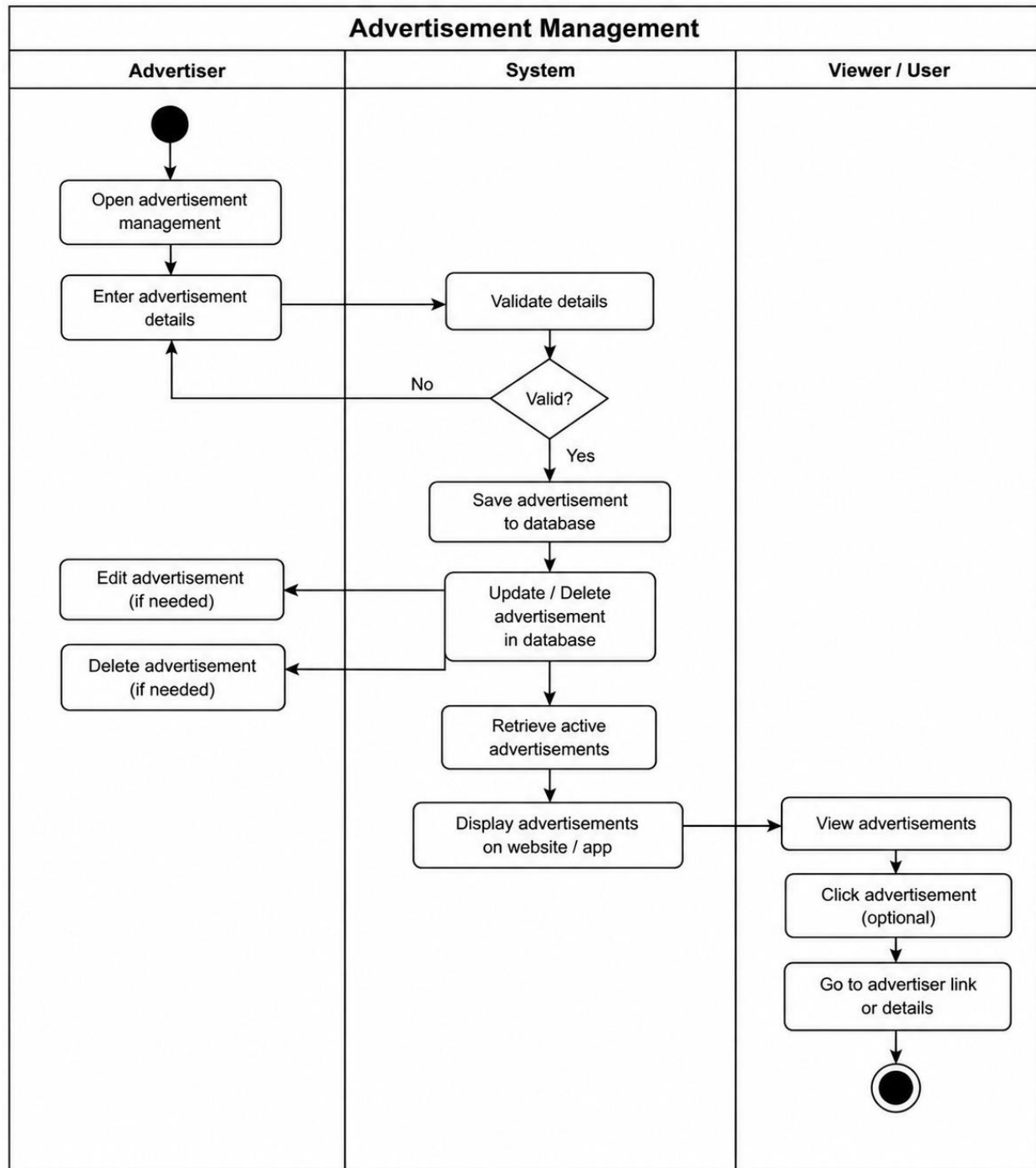
4.1 UC-01: Announcement & Event System – De Silva L. C. A. (IT25101841)



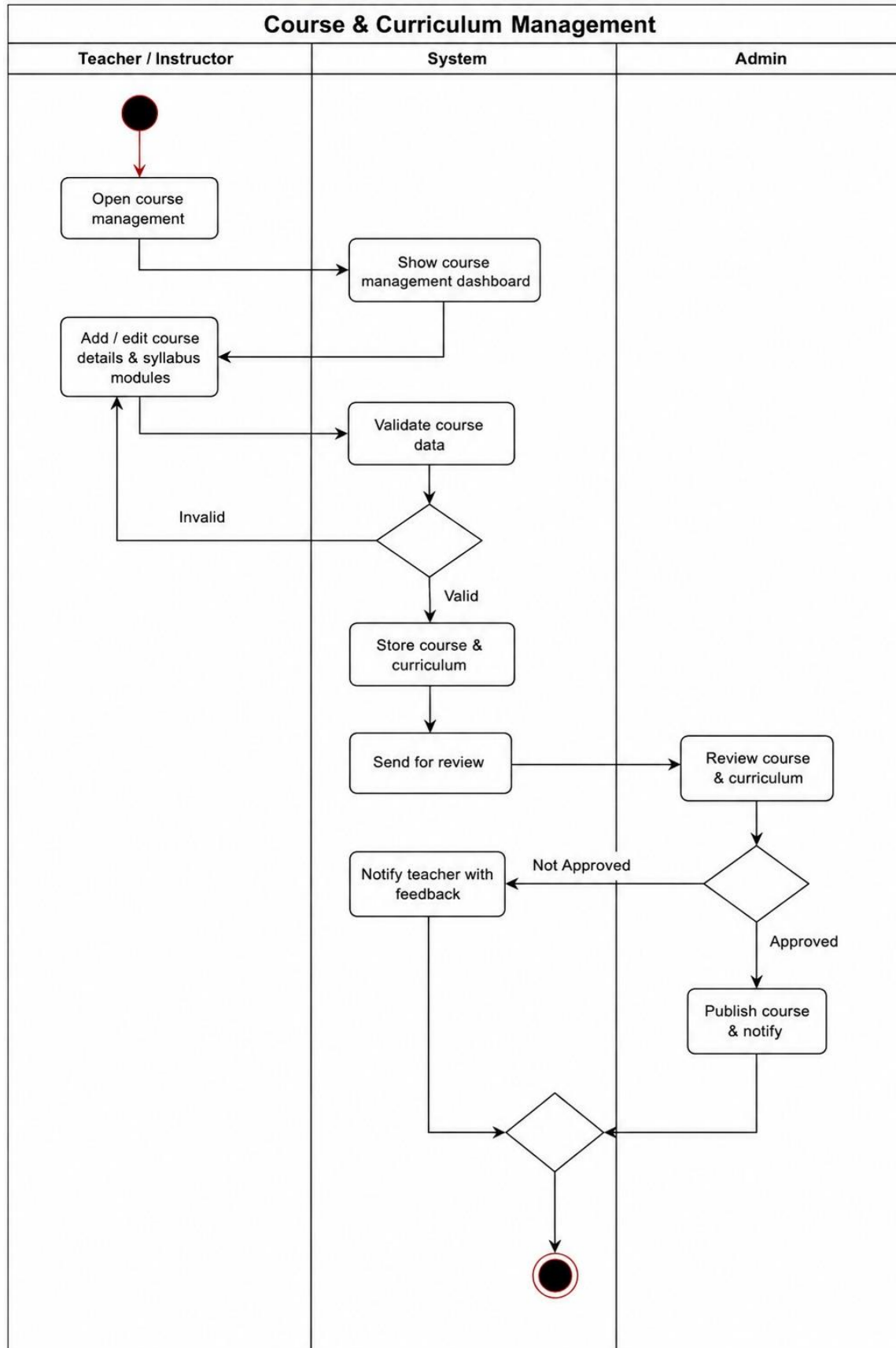
4.2 UC-02: Ticketing System - Munasinghe M. H. V. S. (IT25102695)



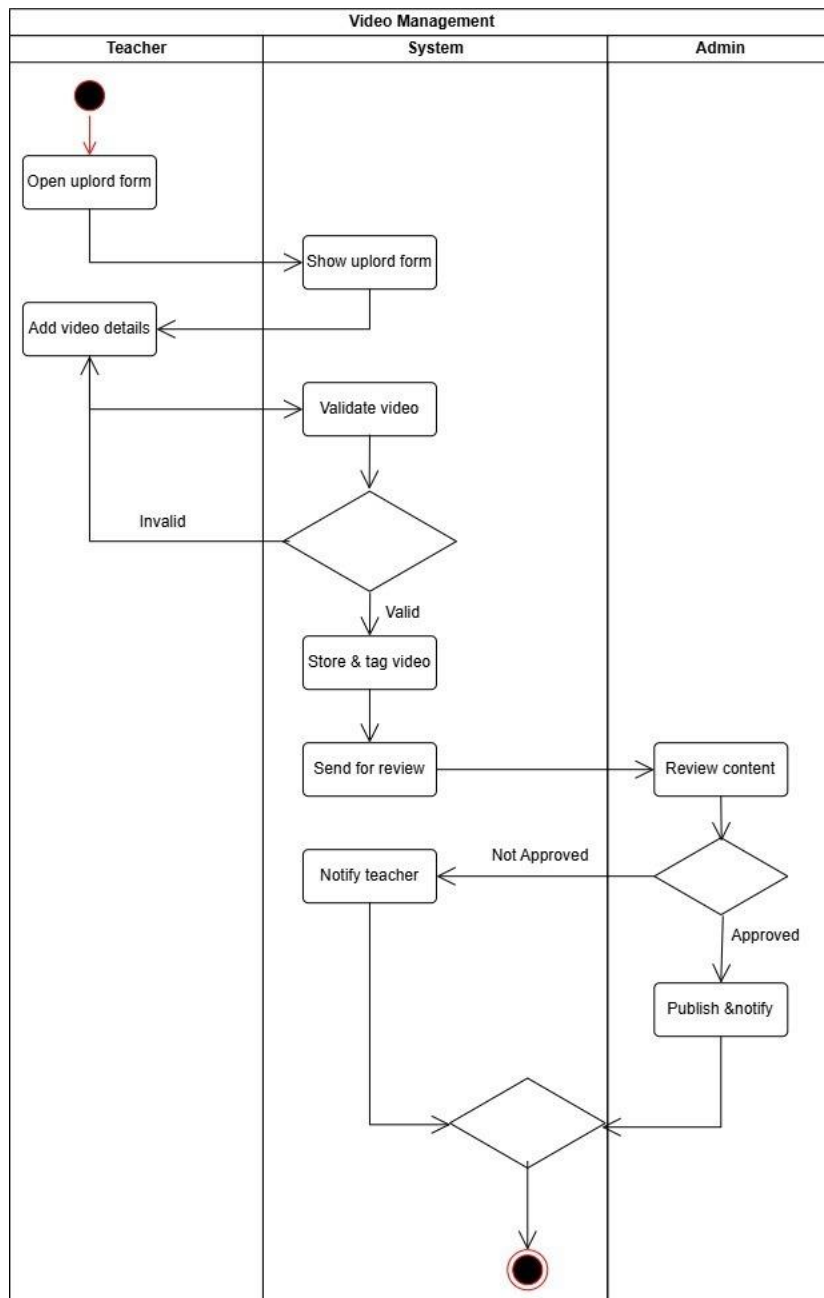
4.3 UC-03: Advertisement System - Bandara W. G. M. D. (IT25103702)



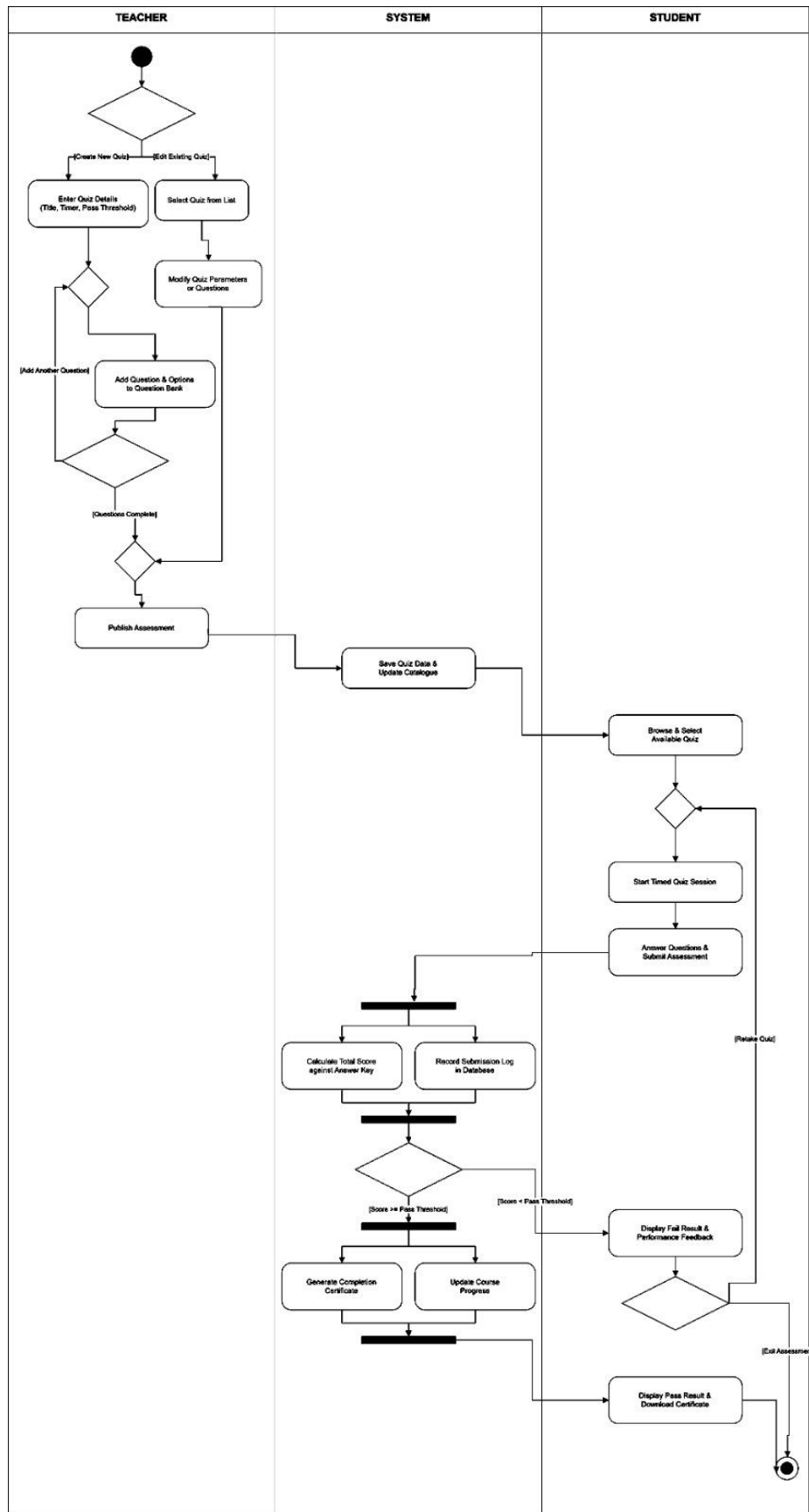
4.4 UC-04: Course & Curriculum Management – Withana T. T. (IT25101767)



4.5 UC-05: Video Management – Wickramathunga P. P. (IT25103673)



4.6 UC-06: Quiz and Assessment – Adikari A. M. T. D. (IT25100859)



5 Ethical Considerations

5.1 UC-01: Announcement & Event System (Assigned to: De Silva L. C. A.)

- **Information Accuracy & Anti-Misinformation:** Broadcast notices and live event notifications directly affect academic schedules. To prevent false broadcasts, only authenticated Teachers and Admins have publishing rights.
- **User Alert Consent:** Automated event notification dispatching adheres to user preference settings on the dashboard to avoid notification spam.

5.2 UC-02: Support Ticketing System (Assigned to: Munasinghe M. H. V. S.)

- **User Data Privacy & Confidentiality:** Support tickets submitted by Students or Teachers contain sensitive technical details and personal issue logs. Enforced **Role-Based Access Control (RBAC)** ensures ticket histories are strictly confidential and visible only to the submitter and authorized Operations Managers.
- **File Attachment Security:** Screenshots and files attached to tickets undergo file-type validation (blocking executable scripts) and encryption at rest to prevent security risks.

5.3 UC-03: Advertisement Promotions (Assigned to: Bandara W. G. M. D.)

- **Truthful Advertising:** Promotional banners are strictly restricted to internal educational offerings and featured courses, preventing third-party ad tracking.
- **Automated Expiry Control:** Advertisements feature configured start and end dates in the database, automatically hiding expired promotional offers from users.

5.4 UC-04: Course & Curriculum Management (Assigned to: Withana T. T.)

- **Intellectual Property Safeguards:** Course modules, syllabus documentation, and downloadable resources are secured to protect teacher ownership.
- **Transparent Enrollment Details:** Prerequisites, course outlines, and learning objectives are clearly displayed before enrollment to prevent misleading expectations.

5.5 UC-05: Video Management (Assigned to: Wickramathunga P. P.)

- **Digital Accessibility & Inclusivity:** Video playback features closed captions, chapter markers, and high-contrast player controls in compliance with **WCAG 2.1 AA** standards for learners with disabilities.
- **Automated Moderation Queue:** Uploaded video files pass through banned keyword algorithms and a pre-publishing moderation queue to quarantine harmful or copyright-infringing content.

5.6 UC-06: Quiz & Assessment Subsystem (Assigned to: Adikari A. M. T. D.)

- **Academic Integrity & Unbiased Scoring:** Timed quizzes with server-side automated grading engines eliminate subjective bias and deliver fair results for all students.
- **Assessment Data Protection:** Question banks, student attempt histories, and generated certificates are protected against score tampering and unauthorized access.